

3.7.2 Populations (A-level only)

Content	Opportunities for skills development
Species exist as one or more populations. A population as a group of organisms of the same species occupying a particular space at a particular time that can potentially interbreed. The concepts of gene pool and allele frequency. The Hardy–Weinberg principle provides a mathematical model, which predicts that allele frequencies will not change from generation to generation. The conditions under which the principle applies. The frequency of alleles, genotypes and phenotypes in a population can be calculated using the Hardy–Weinberg equation: $p^2 + 2pq + q^2 = 1$ where p is the frequency of one (usually the dominant) allele and q is the frequency of the other (usually recessive) allele of the gene.	AT k Students could collect data about the frequency of observable phenotypes within a single population. MS 2.4 Students could calculate allele, genotype and phenotype frequencies from appropriate data using the Hardy–Weinberg equation.